L27-L28 Power Series Solutions to Linear DEQs

Section 8.3

Consider the DEQ

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0 \qquad (1)$$

and its standard form

always say

$$y'' + P(x)y' + Q(x)y = 0 \ , \qquad y = \sum_{n=0}^{\infty} a_n x^n$$

where $P(x) = \dfrac{a_1(x)}{a_2(x)}$ and $Q(x) = \dfrac{a_0(x)}{a_2(x)}$.

Def. A point x_0 is called an **ordinary point** of eq.(1) if both $P(x)$ and $Q(x)$ are analytic at x_0.

If x_0 is not an ordinary point, it is called a **singular point** of the equation.

ex. Classify $x = 0$ as an ordinary point or a singular point of the following equations:

1. $\dfrac{1}{x^2}y'' - \dfrac{2}{x}y' + y = 0 \Rightarrow y'' - 2x\,y' + x^2 y = 0$

   $P = -2x$ yes $x=0$ is ordinary.
   $Q = x^2$

2. $y'' + \cos x y' - e^x y = 0$

   $P = \cos(x)$ yes $x=0$ is ordinary
   $Q = -e^x$

3. $x^2 y'' + 4xy' + xy = 0 \Rightarrow y'' + \dfrac{4}{x}y' + \dfrac{1}{x}y = 0$ $\quad P = \dfrac{4}{x}$, $Q = \dfrac{1}{x}$ $\boxed{x=0 \text{ singular}}$

ex. Determine all the singular points of

$$xy'' + \frac{x}{1-x}y' + (\sin x)y = 0$$

standard form

$$\Rightarrow \quad y'' + \frac{1}{1-x}y' + \frac{\sin(x)}{x}y = 0$$

$P(x) = \frac{1}{1-x}$ not analytic at $x=1$

$$Q(x) = \frac{\sin(x)}{x} = \frac{\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!}}{x} = \frac{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots\right)}{x}$$

$$= 1 - \frac{x^2}{3!} + \frac{x^4}{5!} \cdots$$

$$= \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n+1)!}$$

So the only singular point is $x=1$.

Power series method about an ordinary point

To solve $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ (Assume y is analytic at x_0):

1. Let $\sum_{n=0}^{\infty} a_n (x - x_0)^n$
2. Find y' and y'', and substitute them into the DEQ.
3. Find the recurrence relation and solve for a_n.
4. Apply any given initial conditions if applicable.

<u>**ex.**</u> Find a power series solution about $x = 0$ to

$$y' - \underbrace{2x}y = 0.$$

ordinary

1. $y = \sum_{n=0}^{\infty} a_n x^n \rightarrow y' = \sum_{n=1}^{\infty} a_n n x^{n-1}$

2.

3. $y' - 2xy = 0 \Rightarrow \sum_{n=1}^{\infty} a_n n x^{n-1} - 2x\left(\sum_{n=0}^{\infty} a_n x^n\right) = 0$

$$\Rightarrow \sum_{\underbrace{n=1}_{\downarrow 1}}^{\infty} \underbrace{a_n n x^{n-1}}_{\uparrow 1} + \sum_{\underbrace{n=0}_{\uparrow 1}}^{\infty} \underbrace{-2a_n x^{n+1}}_{\downarrow 1} = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} a_{n+1}(n+1)x^n + \sum_{n=1}^{\infty} -2a_{n-1}x^n = 0$$

$$a_1 \cdot 1 \cdot x^0 + \sum_{n=1}^{\infty} a_{n+1}(n+1)x^n + \sum_{n=1}^{\infty} -2a_{n-1}x^n = 0$$

$$\Rightarrow \quad a_1 + \sum_{n=1}^{\infty} \left[a_{n+1}(n+1) - 2a_{n-1} \right] x^n = 0$$

$$\Rightarrow \quad \begin{cases} a_1 = 0 & (n=0) \\ a_{n+1}(n+1) - 2a_{n-1} = 0, & n \geq 1 \end{cases}$$

$$\Rightarrow \quad \boxed{a_{n+1} = \frac{2a_{n-1}}{n+1}, \quad n \geq 1}$$

recurrence relation.

$n=0$: $a_1 = 0$

$n=1$: $a_2 = \dfrac{2a_0}{2} = a_0 = \dfrac{2}{2} a_0 = \dfrac{1}{1} a_0 = \dfrac{1}{1!} a_0$

$n=2$: $a_3 = \dfrac{2a_1}{3} = \dfrac{2}{3} \cdot 0 = 0$

$n=3$: $a_4 = \dfrac{2a_2}{4} = \dfrac{1}{2} a_0 = \dfrac{2}{4} \cdot \dfrac{2}{2} a_0 = \dfrac{1}{2} a_0 = \dfrac{1}{2!} a_0$

$n=4$: $a_5 = \dfrac{2a_3}{5} = 0$

$n=5$: $a_6 = \dfrac{2a_4}{6} = \dfrac{2}{6} \cdot \dfrac{2}{4} \cdot \dfrac{2}{2} a_0 = \dfrac{1}{3 \cdot 2 \cdot 1} a_0 = \dfrac{1}{6} a_0 = \dfrac{1}{3!} a_0$

$\boxed{a_{odd} = 0}$ $\qquad$ $\boxed{a_{2n} = \dfrac{1}{n!} a_0}$

$n=7$: $a_8 = \dfrac{2a_6}{8} = \dfrac{2}{8} \cdot \dfrac{2}{6} \cdot \dfrac{2}{4} \cdot \dfrac{2}{2} a_0 = \dfrac{1}{4!} a_0$

$$\text{So} \quad y = \sum_{n=0}^{\infty} a_n x^n = \underbrace{\sum_{n=0}^{\infty} a_{2n} x^{2n}}_{\text{even index}} + \underbrace{\cancel{\sum_{n=0}^{\infty} a_{2n+1} x^{2n+1}}}_{\text{odd}}^{\,0}$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} a_0 x^{2n} = a_0 \sum_{n=0}^{\infty} \frac{1}{n!} \underbrace{x^{2n}}_{(x^2)^n} = a_0 e^{x^2}$$

ex. Find a power series solution about $x=0$ to

$$y''+y=0. \qquad \Rightarrow y = c_1\cos(x) + c_2\sin(x)$$

$$y=\sum_{n=0}^{\infty} a_n x^n, \quad y' = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$$

$$y''+y = \sum_{\substack{n=2\\ \downarrow 2}}^{\infty} \underbrace{n(n-1)a_n x^{n-2}}_{\uparrow 2} + \sum_{n=0}^{\infty} a_n x^n = 0 \quad \checkmark$$

(1)

$$\Rightarrow \sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty}\Big[\underbrace{(n+2)(n+1)a_{n+2} + a_n}_{=0}\Big]x^n = 0$$

(2)

$$\Rightarrow (n+2)(n+1)a_{n+2} + a_n = 0, \quad n \geq 0$$

shift 2

$$\Rightarrow \boxed{a_{n+2} = \frac{-a_n}{(n+2)(n+1)}, \quad n \geq 0}$$

recurrence relation

$y''+y=0$ second order

two constants a_0, a_1

(3)

$$n=0: \quad a_2 = \frac{-a_0}{2\cdot 1} = -1\cdot\frac{1}{2\cdot 1}a_0 = \frac{-1}{2!}a_0$$

$$n=1: \quad a_3 = \frac{-a_1}{3\cdot 2} = -1\cdot\frac{1}{3\cdot 2}a_1 = \frac{-1}{3!}a_1$$

$$n=2: \quad a_4 = \frac{-a_2}{4\cdot 3} = -1\cdot\frac{1}{4\cdot 3}\cdot(-1)\frac{1}{2\cdot 1}a_0 = \frac{(-1)^2}{4!}a_0$$

$$n=3: \quad a_5 = \frac{-a_3}{5\cdot 4} = -1\cdot\frac{1}{5\cdot 4}\cdot(-1)\frac{1}{3\cdot 2}a_1 = \frac{(-1)^2}{5!}a_1$$

$n=4$: $a_6 = \dfrac{-a_4}{6\cdot 5} = \dfrac{-1}{6\cdot 5}\,\dfrac{(-1)^2}{4!}a_0 = \dfrac{(-1)^3}{6!}a_0$

$n=5$: $a_7 = \dfrac{-a_5}{7\cdot 6} = \dfrac{-1}{7\cdot 6}\cdot\dfrac{(-1)^2}{5!}a_1 = \dfrac{(-1)^3}{7!}a_1$

$$a_n = \begin{cases} \dfrac{(-1)^m}{(2m)!}a_0 = a_{2m} & n \text{ even}, \quad n=2m \\[2ex] \dfrac{(-1)^m}{(2m+1)!}a_1 = a_{2m+1} & n \text{ odd}, \quad n=2m+1 \end{cases}$$

(4) So $y = \displaystyle\sum_{n=0}^{\infty} a_n x^n = \underbrace{\sum_{n=0}^{\infty} a_{2n}x^{2n}}_{\text{evens}} + \underbrace{\sum_{n=0}^{\infty} a_{2n+1}x^{2n+1}}_{\text{odds}}$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}a_0 x^{2n} + \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}a_1 x^{2n+1}$$

$$= a_0 \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n)!}x^{2n} + a_1\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}x^{2n+1}$$

$$= a_0\cos(x) + a_1\sin(x)$$

ex. Find a general solution to

$$2y'' + xy' + y = 0$$

↑variable coefficient

in the form of a power series about $x = 0$.

let $y = \sum_{n=0}^{\infty} a_n x^n$

$$2y'' + xy' + y = 2\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + x\sum_{n=1}^{\infty} n a_n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{\substack{n=2 \\ \downarrow 2}}^{\infty} \underbrace{2(n)(n-1) a_n x^{n-2}}_{\uparrow 2} + \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} 2(n+2)(n+1) a_{n+2} x^n + \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

separate $n=0$ terms:

$$\underbrace{\left[2(2)(1)a_2 x^0 + a_0 x^0\right]}_{n=0} + \sum_{n=1}^{\infty} \left[2(n+2)(n+1)a_{n+2} + \underline{n a_n + a_n}\right] x^n = 0$$

$$\Rightarrow \begin{cases} 4a_2 + a_0 = 0 & (n=0) \\ 2(n+2)(n+1)a_{n+2} + (n+1)a_n = 0, & n \geq 1 \end{cases}$$

$$\Rightarrow \boxed{a_{n+2} = \frac{-\cancel{(n+1)}a_n}{2(n+2)\cancel{(n+1)}} = \frac{-a_n}{2(n+2)}, \quad n \geq 1}$$

recurrence relation

$n=0: \quad 4a_2 + a_0 = 0 \Rightarrow a_2 = -\frac{1}{4}a_0$

$n=1: \quad a_3 = \frac{-a_1}{2(3)} = \frac{-1}{2\cdot 3}a_1$

$n=2: \quad a_4 = \frac{-a_2}{2(4)} = \frac{-1}{2\cdot 4}\cdot\frac{-1}{4}a_0$

$n=3: \quad a_5 = \frac{-a_3}{2\cdot 5} = \frac{-1}{2\cdot 5}\cdot\frac{-1}{2\cdot 3}a_1$

$n=4: \quad a_6 = \frac{-a_4}{2\cdot 6} = \frac{-1}{2\cdot 6}\,\frac{-1}{2\cdot 4}\,\frac{-1}{\underset{2\cdot 2}{4}}a_0 = \frac{(-1)^3}{2^3\,\underbrace{2\cdot 4\cdot 6}_{6!!}}a_0$

$= \frac{(-1)^3}{2^3\cdot \underbrace{2^3(1\cdot 2\cdot 3)}_{2\cdot 4\cdot 6}}a_0$

$= \frac{(-1)^3}{2^6\,3!}a_0$

$n=5: \quad a_7 = \frac{-a_5}{2\cdot 7} = \frac{-1}{2\cdot 7}\cdot\frac{-1}{2\cdot 5}\cdot\frac{-1}{2\cdot 3}a_1$

$= \frac{(-1)^3}{2^3\cdot 3\cdot 5\cdot 7}a_1$

in general $a_n = \begin{cases} \dfrac{(-1)^m}{2^{2m}\,m!}a_0, & n=2m \text{ (even)} \\ \dfrac{(-1)^m}{2^m\cdot\underbrace{3\cdot 5\cdot 7\cdots(2m+1)}_{(2m+1)!!}}a_1, & n=2m+1 \end{cases}$

So $y = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_{2n}x^{2n} + \sum_{n=0}^{\infty} a_{2n+1}x^{2n+1}$

$= \underbrace{\sum_{n=0}^{\infty}\frac{(-1)^n}{2^{2n}n!}a_0x^{2n}} + \sum_{n=0}^{\infty}\frac{(-1)^n}{2^n\,3\cdot 5\cdot 7\cdots(2n+1)}a_1x^{2n+1}$

$a_0\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\left(\frac{x}{2}\right)^{2n} = a_0\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\left(\frac{x^2}{4}\right)^n = a_0e^{-x^2/4}$

$y = a_0e^{-x^2/4} + a_1\sum_{n=0}^{\infty}\frac{(-1)^n}{2^n\cdot 3\cdot 5\cdot 7\cdots(2n+1)}x^{2n+1}$

so $y_1 = e^{-x^2/4}$ is a solution to $2y'' + xy' + y = 0$

$\Rightarrow y'' + \underbrace{\tfrac{x}{2}}_{p(x) = \frac{x}{2}} y' + \tfrac{1}{2} y = 0$

Reduction of Order : y_1 soln , y_2

$$y_2 = y_1 \int \frac{e^{-\int p(x)\,dx}}{y_1^{\,2}}\,dx$$

$$\Rightarrow y_2 = e^{-x^2/4} \int \frac{e^{-\int \frac{x}{2} dx}}{(e^{-x^2/4})^2}\,dx = e^{-x^2/4} \int \frac{\cancel{e^{-x^2/4}}}{(e^{-x^2/4})^{\cancel{2}}}\,dx$$

$$= e^{-x^2/4} \int e^{x^2/4}\,dx$$

error function : $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,dt$

ex. Find at least the first four nonzero terms in a power series expansion about $x=0$ for the solution to the IVP:

$$y''+(x-2)y'-y=0\,,\quad y(0)=-1\,,\ y'(0)=0$$

$$y=\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$$

$$y(0)=a_0=-1$$

$$y'=\sum_{n=1}^{\infty} n a_n x^{n-1} = a_1 + 2a_2 x + \dots$$

$$y'(0)=a_1=0$$

$$y''+(x-2)y'-y=\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + (x-2)\sum_{n=1}^{\infty} n a_n x^{n-1} - \sum_{n=0}^{\infty} a_n x^n = 0$$

$$=\sum_{\underbrace{n=2}_{\downarrow 2}}^{\infty} \underbrace{n(n-1)a_n x^{n-2}}_{\uparrow 2} + \sum_{n=1}^{\infty} n a_n x^n - \sum_{\underbrace{n=1}_{\downarrow 1}}^{\infty} \underbrace{2n a_n x^{n-1}}_{\uparrow 1} - \sum_{n=0}^{\infty} a_n x^n = 0$$

$$=\sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n + \sum_{n=1}^{\infty} n a_n x^n - \sum_{n=0}^{\infty} 2(n+1)a_{n+1}x^n - \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\overbrace{\left[2a_2-2a_1-a_0\right]}^{n=0} + \sum_{n=1}^{\infty}\left[(n+2)(n+1)a_{n+2} + n a_n - 2(n+1)a_{n+1} - a_n\right]x^n = 0$$

$$\Rightarrow \begin{cases} 2a_2-2a_1-a_0=0 & (n=0) \\ (n+2)(n+1)a_{n+2}-2(n+1)a_{n+1}+(n-1)a_n=0, & n\geq 1 \end{cases}$$

$$\Rightarrow \quad a_{n+2} = \frac{2(n+1)a_{n+1} - (n-1)a_n}{(n+2)(n+1)}, \quad n \geq 1$$

- $\boxed{a_0 = -1}$
- $a_1 = 0$

- $2a_2 - 2a_1 - a_0 = 0 \Rightarrow 2a_2 = 2a_1 + a_0 = 2\cdot 0 + (-1)$

$$\Rightarrow \boxed{a_2 = -\frac{1}{2}}$$

- $n=1, \quad a_3 = \dfrac{2(2)a_2 - (1-1)a_1}{3\cdot 2} = \dfrac{4(-\frac{1}{2}) - 0}{6} = -\dfrac{1}{3}$

$$\Rightarrow \boxed{a_3 = -\frac{1}{3}}$$

- $n=2, \quad a_4 = \dfrac{2(3)a_3 - (2-1)a_2}{4\cdot 3} = \dfrac{-2 + \frac{1}{2}}{12} = -\dfrac{1}{8}$

$$\Rightarrow \boxed{a_4 = -\frac{1}{8}}$$

so $\quad y = \displaystyle\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots$

$$= -1 + 0\cdot x + \left(-\tfrac{1}{2}\right)x^2 + \left(-\tfrac{1}{3}\right)x^3 + \left(-\tfrac{1}{8}\right)x^4 + \cdots$$

$$= \boxed{-1 - \tfrac{1}{2}x^2 - \tfrac{1}{3}x^3 - \tfrac{1}{8}x^4} + \cdots$$